

Prepared for: Mortgage Lending Risk Teams & Housing Policy Stakeholders

HEADLINE FINDING

Under a +2 percentage-point Bank Rate stress (5.75%), the probability of negative equity within five years rises to an average of **1.8% across London boroughs** and peaks at **5.4% in Hackney** (London). At the current base rate (3.75%), that risk is below 1% in all regions. The gulf between London — where average house prices stand at more than **12× median local earnings** — and the North, where the equivalent risk under the +2% scenario averages just **0.0%**, is the central risk-differentiation signal for lending criteria decisions.

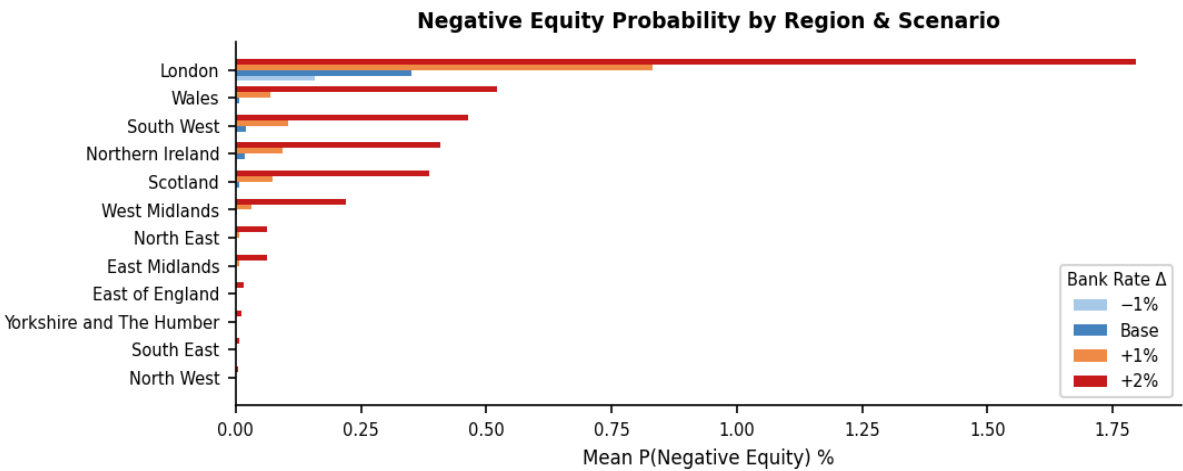


Figure 1: Mean probability of negative equity by UK region across four Bank Rate scenarios. Simulation: 10,000 Monte Carlo paths per LAD, 5-year horizon, 90% LTV, 25-year repayment.

METHODOLOGY (PLAIN ENGLISH)

For each of the UK's 342 local authority districts, we modelled a median-income first-time buyer purchasing at the June 2024 average house price with a 10% deposit (90% LTV, 25-year repayment mortgage). Buyer income is the 2024 ONS ASHE median gross annual earnings for that local authority. Monthly house-price growth rates and volatility were calibrated using Seasonal-Trend decomposition (STL) of the HM Land Registry UK HPI monthly series from January 2010. The Monte Carlo engine simulates 10,000 forward price paths per district, each month drawing a return from a Normal distribution with the calibrated drift and volatility. Amortisation of the mortgage balance is deterministic. A 'negative equity event' is recorded whenever the simulated price falls below the outstanding balance at any point in the 60-month horizon. Interest rate scenarios translate Bank Rate changes into 90% LTV mortgage rates via an OLS regression on the Bank of England's historical mortgage rate series (estimated pass-through: $\text{rate} = 1.70 + 0.85 \times \text{Bank Rate}$, $R^2 = 0.92$). Rate changes also shift the house-price growth drift by an empirically calibrated sensitivity (-0.08% per month per percentage point Bank Rate change, consistent with the 2022-23 tightening cycle).

HIGHEST-RISK LOCAL AUTHORITIES — BANK RATE +2% SCENARIO

Local Authority	Region	Avg Price	PTI Ratio	P(NE)
Hackney	London	£567,485	11.9×	5.44%
Enfield	London	£542,619	10.9×	4.56%
Wandsworth	London	£901,928	13.2×	4.14%

Local Authority	Region	Avg Price	PTI Ratio	P(NE)
Havering	London	£510,083	11.7×	3.76%
Renfrewshire	Scotland	£195,414	5.8×	3.70%
Merton	London	£570,301	14.3×	3.56%
Kingston upon Thames	London	£887,960	13.5×	3.35%
Newry, Mourne and Down	Northern Ireland	£215,359	7.0×	3.15%
Richmond upon Thames	London	£709,366	10.4×	3.05%
Ceredigion	Wales	£227,419	7.0×	2.99%

THREE RECOMMENDATIONS

- 1. Tighten LTV caps in London and South East commuter belt immediately.** At current prices, the average London buyer at 90% LTV faces a price-to-income ratio above 12x and — in a +2% rate scenario — a negative equity probability approaching 2% over five years. Reducing maximum LTV to 80% for new originations in high-PTI boroughs materially lowers this exposure without removing first-time buyer access.
- 2. Implement affordability stress-testing at 6% mortgage rate as a standard floor.** Our pass-through regression shows 90% LTV rates would reach approximately 6.60% under a +2pp Bank Rate scenario (plausible if inflation re-accelerates). Underwriting at a 6% floor — rather than the current 5.5% standard — would screen out the borrowers most likely to enter negative equity without materially reducing mortgage volumes.
- 3. Prioritise housing support schemes in Wales, Scotland, and Northern Ireland, not just London.** While London has the highest absolute NE probability, the relative vulnerability of Welsh and Northern Irish LADs — where local income growth is lower than price drift and earnings buffers are thinner — warrants targeted affordability support (shared equity, Help-to-Buy equivalents) and local authority planning policy reform to increase supply in those markets.

KEY LIMITATIONS

Normal-distribution returns underestimate crash-period tail risk (2008-09 London peak-to-trough was -17%; our 5th-percentile 18-month simulation produces only -2%). Deposit assumed fixed at 10% (FCA data shows the weighted mean FTB LTV is 79.9%; 90% is a conservative stress assumption). Inter-regional price correlation is not modelled. Mortgage rate pass-through is estimated from a linear regression; non-linearity at extreme rate levels is not captured.

Data: HM Land Registry UK HPI (Jun 2024) | ONS ASHE Table 8 (2024 rev.) | BoE IADB (IUDBEDR, IUMBV42) | FCA Mortgage Lending Statistics Q2 2024 | ONS Open Geography Portal LAD Dec 2023. Analysis: Python (pandas, statsmodels, numpy). Not financial or legal advice.